INORGANIC CHEMISTRY CUMULATIVE EXAM

October 22, 1994

Answer any **6** of the following questions:

1. Sketch an octahedral complex with one tridentate, one bidentate (chelate), and one monodentate ligand. Is it optically active? Show how you came to this conclusion.

2. Sketch four isomers possible for the "twisted cube" (square antiprism) complex UF_6Cl_2.

3. Give a way that the chelate ethylenediamine could be chemically modified so that it could be used to determine the absolute configuration of a tris chelate complex using X-ray structural methods.

4. Write a chemical reaction which would illustrate the chelate effect.

5. Describe how the ligand SO_2 might show linkage isomerism and give the factors which would influence its mode of bonding.

6. Design and sketch a logical cryptand ligand which uses S atoms as the donor atoms. Show all atoms and bonds.

7. Why do substitution reactions of a square planar complex typically result in the new ligand occupying exactly the same site (position) formerly occupied by the ligand which was displaced?

8. The rate law for the reaction of PtL_4^{+2} with X^- was found to be: Rate = $k[PtL_4^{+2}]$. Describe how this reaction could possibly be associative even though the rate law shows a dependency on only the complex concentration.

9. Describe the difference in the factors which are the main contributors to the activation energy for inner and outer sphere redox mechanisms.